

Basil C K

*AI & Data Science Graduate | Python Backend | Machine Learning |
Computer Vision*

basilck618@gmail.com | +91 8111974510 | Portfolio: basilbaasi.github.io | LinkedIn | GitHub

Thrissur, Kerala, India

Profile

B.Tech graduate in Artificial Intelligence & Data Science with strong skills in Python, Machine Learning, Computer Vision, and Backend Development. Hands-on experience building end-to-end ML pipelines, REST APIs, and AI-driven applications. Proficient in data preprocessing, model development, and integrating machine learning models into real-world systems. Familiar with SQL, version control (Git), and software development fundamentals including operating systems and networking. Seeking entry-level roles in AI, Backend Development, and Machine Learning Engineering.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

JCT College of Engineering and Technology, Anna University

2022 – 2026

CGPA: 8.2 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Machine Learning

TECHNICAL SKILLS

Programming & Backend:

Python (automation, backend logic, ML pipelines), Flask, Django, Django REST Framework, REST APIs, CRUD, API Development

Machine Learning & Computer Vision:

CNN architectures, YOLOv8 object detection, OpenCV image processing, Computer Vision, model training & evaluation, NLP fundamentals, Generative AI concepts

Data & ML Foundations:

Pandas, NumPy, supervised learning fundamentals, SQL (queries, joins, basic schema design)

Systems & Networking:

Operating Systems fundamentals, HTTP/ REST architecture, client-server systems, TCP/ IP basics

Cloud & Platforms:

Basic exposure to OCI / AWS fundamentals

Tools & Development:

Git (branching, commits, workflows), Jupyter Notebook, VS Code

AI & LLM Exposure:

LLM Integration, Chatbot Development, Prompt Engineering, conversational AI workflows

Exposure to:

Linux (CLI, permissions, processes), Docker (containerization basics), CI/ CD pipelines (conceptual understanding), ML model deployment workflows

INTERNSHIP EXPERIENCE

Data Analytics Virtual Experience – Accenture (Forage)

- Performed data cleaning, exploratory analysis, and visualization on structured datasets.
- Identified user engagement trends and generated insights for a simulated business scenario.
- Presented findings and recommendations in a structured report format.

PROJECT EXPERIENCE

Vexilon – AI-Powered Chrome Assistant

Tech Stack: Python, Flask, REST APIs, LLM APIs, Chrome Extension, Git

- Built a Chrome extension integrated with a Flask backend enabling real-time AI-based conversational interaction
- Designed RESTful APIs for request handling and response streaming
- Integrated external LLM APIs (Gemma, Ollama) using OpenAI-compatible formats
- Implemented modular backend structure with session-based interaction workflows

Entity Management API – Django REST Backend

Tech Stack: Python, Django, Django REST Framework, Swagger

- Developed a RESTful API for managing structured entities with full CRUD functionality
- Implemented data validation, relational mappings, and modular API design
- Documented endpoints using Swagger for usability and testing
- Designed scalable backend structure following clean architecture principles

Hybrid CNN-YOLO Car Damage Detection

Tech Stack: Python, YOLOv8, TensorFlow/Keras, PyTorch, OpenCV, Flask, HTML/CSS

- Built a computer vision pipeline combining YOLOv8 detection and CNN classification for vehicle damage analysis
- Designed multi-stage inference system for damage localization and severity prediction
- Achieved mAP@0.5 of 0.659 and validation accuracy of 54.4% on custom dataset
- Developed a Flask-based interface for real-time image upload and prediction

Vocalix – AI Voice Assistant

Tech Stack: Python, Speech Recognition, LLM APIs, Automation Libraries

- Developed a voice-controlled assistant for executing system-level tasks
- Integrated LLM APIs for conversational responses
- Implemented speech-to-command processing and automation workflows
- Designed structured interaction pipeline for real-time voice-based control

ADDITIONAL LEARNING & CERTIFICATIONS

- **IBM Data Science** – Selected Coursework (Python, Data Analysis)
- **Oracle Cloud Infrastructure 2023** – AI Foundations Associate
- **Google AI Essentials**

LANGUAGES

English | Malayalam | Tamil